

Grade 8
Unit 5
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

1. Find the horizontal distance between the points $(-1, 8)$ and $(-7, 5)$.

$$|-7 - (-1)| = |-7 + 1| = |-6| = 6 \text{ units}$$

2. To find the horizontal distance on the coordinate plane, find the between the

- ☐ y - coordinates.
☒ x - coordinates.

- ☒ difference
☐ sum

3. Find the vertical distance between the points $(-1, 8)$ and $(-7, 5)$.

$$|8 - 5| = |3| = 3 \text{ units}$$

4. To find the vertical distance on the coordinate plane, find the between the

- ☒ y - coordinates.
☐ x - coordinates.

- ☒ difference
☐ sum

Use Triangle *HUS* for questions 5 – 9.

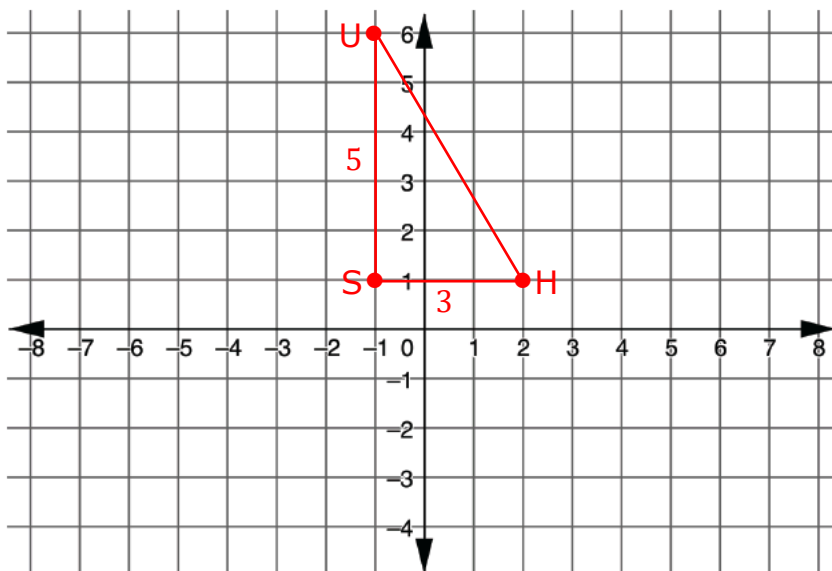
Triangle *HUS* is defined by the following vertices in the coordinate plane.

Point *H* $(2, 1)$

Point *U* $(-1, 6)$

Point *S* $(-1, 1)$

5. Graph Triangle *HUS*.



6. Find the length of the horizontal leg of Triangle HUS . Label the length of the leg on the graph.

3 units

7. Find the length of the vertical leg of Triangle HUS . Label the length of the leg on the graph.

5 units

8. Use the Pythagorean Theorem to find the length of $\overline{HU}$. Round to the nearest tenth.

$$3^2 + 5^2 = c^2$$

$$9 + 25 = c^2$$

$$\sqrt{9 + 25} = c = \sqrt{34} \approx 5.8$$

9. Find the perimeter of Triangle HUS .

$$P = 3 + 5 + 5.8 = 13.8 \text{ units}$$

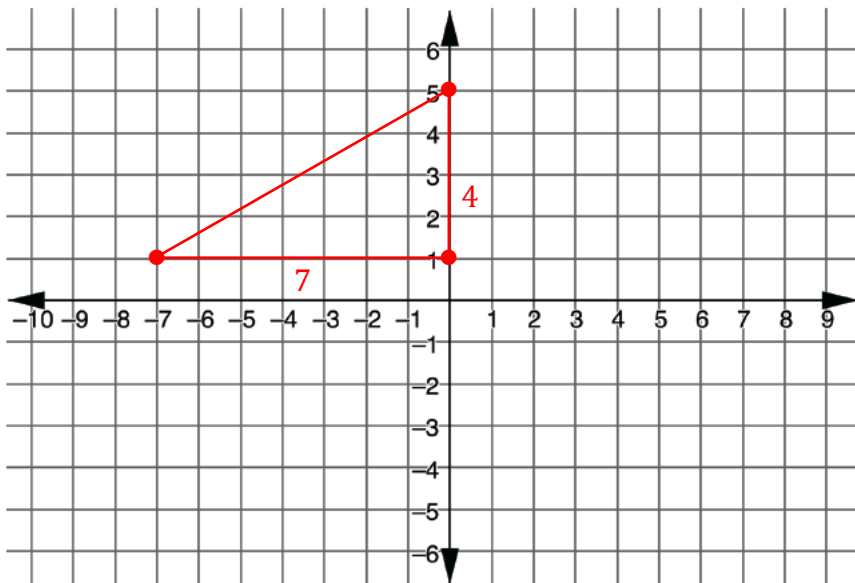
10. Triangle ABC is defined by the vertices.

$A (-7, 1)$

$B (0, 5)$

$C (0, 1)$

Find the perimeter of Triangle PGF . Round to the nearest tenth, if needed.



$$c^2 = 4^2 + 7^2$$

$$c^2 = 16 + 49$$

$$c^2 = 65$$

$$c = \sqrt{65}$$

$$c \approx 8.1$$

$$P = 4 + 7 + 8.1$$

$$= 19.1 \text{ units}$$